

Engineering Programming II: Random Number Generation

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1 Random variables

Let us start with the definition of a random variable. A **random variable** X is a quantity that can “randomly” take on any of a number of different values. The set of possible values it can assume (i.e., the set of possible **outcomes**) is called the **sample space**, often denoted S . Consider the simple example of flipping a coin. In this case, $S = \{\text{heads}, \text{tails}\}$, so a random variable drawn from this sample space could assume outcomes of either *heads* or *tails*. Now, we must also define a **probability distribution** P over this sample space so that we can determine the likelihoods of the outcomes¹.

2 Drawing from distributions in Python

1. Random integer
2. Choice
3. Uniform
4. Normal
5. Multivariate normal

¹Technically, P is a **measure** on S , meaning it is defined on *subsets* of S , not elements of it. However, it is common to abuse notation and talk about $P(X)$ as opposed to $P(\{X\})$.

3 Uniform distributions over specific domains

3.1 Rectangle

It is very easy to create a uniform distribution over a rectangle (or high-dimensional analogue). For example, if we wanted to sample a uniform distribution over the rectangle $[0, 2] \times [0, 1]$, we can construct points (x_i, y_i) where $x_i \sim U(0, 2)$ and $y_i \sim U(0, 1)$ for all i . See Figure 1 for an illustration of this.

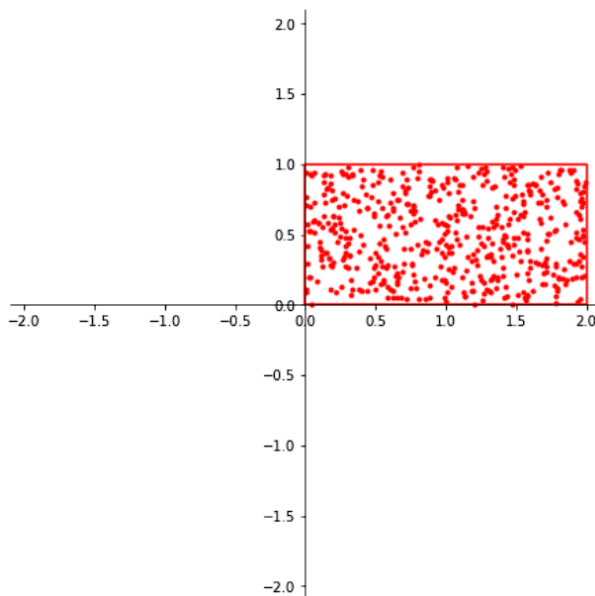


Figure 1: Uniformly sampled points from $[0, 2] \times [0, 1]$.

3.2 Circle

However, generating uniformly sampled points from, say, a circle is slightly more involved. We cannot simply define the bounds on four edges. We must be more clever. The problem can be solved by using *polar coordinates*, i.e., we generate angles θ and radii r for the points. Let us try to uniformly sample points from a circle of radius 1 centered on the origin.

THE WRONG WAY

It should be immediately noted that there is a mistake that is all too easy to make. At first glance, one might think that we can simply draw $\theta_i \sim [0, 2\pi)$ and $r_i \sim [0, 1]$. Doing this causes the distribution to be more dense in the center since it ignores the fact that the “ring’s” of constant r around the circle get longer in circumference as they get further from the center. Thus, a ring of constant r close to the center will have just as many points as a ring far from the center despite the fact that it is shorter. See the left graph in Figure 2 for an illustration.

THE RIGHT WAY

To fix this we simply draw $\theta_i \sim [0, 2\pi)$ and $r_i^2 \sim [0, 1]$ (or, $r_i = \sqrt{\hat{r}_i}$ where $\hat{r}_i \sim [0, 1]$). In other words, we take the square root of the random number we

draw. Doing this accounts for the increasing circumference of the rings of constant r . See the right graph in Figure 2 for an illustration.

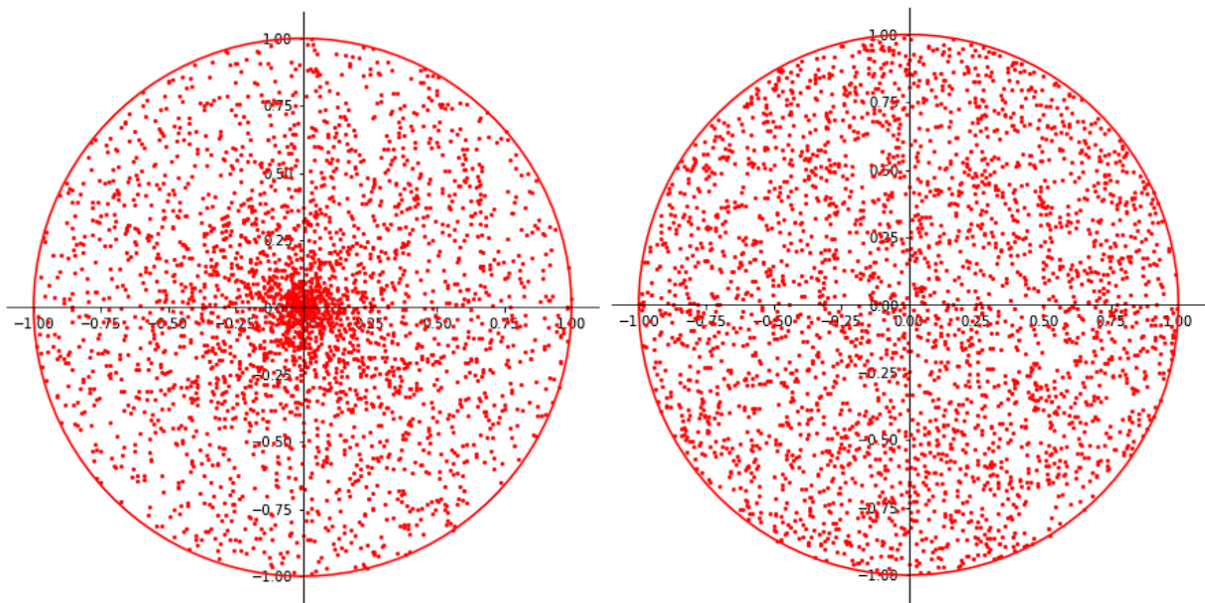


Figure 2: Left: failing to uniformly sample the circle. Right: correctly uniformly sampling the circle.

4 Homework

4.1 Random walk on plane

4.2 Uniform distribution on a cylinder

4.3 Uniform distribution on a sphere

scale by $\cos$

4.4 Random walk on sphere